

HEAVY SAFE HELICOPTER (HSC)

DOCUMENT 00 — COMPREHENSIVE ENGINEERING

BASELINE

Generation 1 | 23-Ton-Class | 7000 kg Capsule

Separable Occupied Survival Capsule + Controlled Rapid Separation + Multi-Mode Recovery

Document Title	HSC Generation 1 — Comprehensive Engineering Baseline
Document Number	HSC-00
Revision	Rev.01-DoubleFont — 23 000 kg / 22 m rotor / 7 000 kg capsule
Status	[PROPOSED] Preliminary Controlled Engineering Baseline
Founder	Amirbahador Atai

Engineering integrity statement: This document defines a controlled conceptual baseline. Numerical values marked TARGET, ASSUMPTION, STUDY, CALC or OPEN are not demonstrated performance, certification results, or flight-proven values.

Revision intent: Establish a controlled Generation 1 study baseline of 23 000 kg design mass, 22.0 m main rotor, 3.6 m tail rotor and 1 000 km range target, with a fully allocated 7 000 kg occupied survival capsule.

1. PURPOSE AND SCOPE

HSC Generation 1 defines the system-level architecture for a 23-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain rather than reliance on a single emergency device.

The baseline covers the aircraft, occupied survival capsule, rapid selective separation system, safety initiation paths, deterministic safety authorization, stabilization, recovery systems, impact attenuation, post-event survival and the interfaces required to verify the complete chain.

The design philosophy is not separate everything whenever the aircraft starts descending. Separation is a controlled emergency mode authorized only when defined conditions are satisfied. The system must account for altitude, airspeed, attitude, angular rates, structural state, fire state, rotor state, available recovery modes, capsule clearance and downstream hazards.

The survival architecture supports multiple emergency outcomes: conventional flight recovery where possible, autorotation where applicable, selective hazard/mass reduction, capsule separation, parachute-assisted descent, energy-absorbing landing and post-landing rescue.

Human emergency authority is preserved. Automatic systems may initiate or recommend a survival sequence only within defined authorization envelopes; the crew must retain an independent means to command, inhibit or terminate automatic action where technically and

physically possible.

This document is an engineering/IP baseline, not a certification approval. Airworthiness, structural substantiation, parachute qualification, release reliability, occupant injury criteria, flight performance and global patent novelty remain open work items.

2. BASELINE STATUS AND CONFIGURATION CONTROL

Status Label	Meaning
[LOCKED CONCEPT]	Architecture retained as baseline design principle; implementation subject to verification.
[REQ]	Top-level requirement proposed for subsequent allocation.
[TARGET]	Numerical or functional development objective; not demonstrated.
[ASSUMPTION / STUDY]	Working engineering assumption for preliminary sizing.
[CALC]	Value calculated from stated assumptions and equations.
[OPEN]	Requires analysis, simulation, test or design decision.

The baseline shall be configuration-controlled. Any change to MTOW, rotor diameter, capsule mass, occupant count, separation sequence, parachute architecture, landing-support geometry or safety-layer independence shall trigger an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

3. GENERATION 1 TOP-LEVEL TECHNICAL BASELINE

Parameter	Value	Status
Maximum Take-Off / Design Mass	23 000 kg	[TARGET]
Occupancy	10–12 persons	[TARGET / STUDY]
Flight architecture	Single main rotor + tail rotor	[REQ / STUDY]
Main rotor diameter	22.0 m	[TARGET]
Main rotor disk area	380.13 m²	[CALC]
Weight-based disk loading	60.5 kg/m²	[CALC]
Tail rotor diameter	3.6 m	[TARGET]
Propulsion	Two turboshaft-class engines	[REQ / STUDY]
Landing gear	8 fixed short mechanical supports under capsule	[TARGET]
Range	1 000 km	[TARGET]
Survival capsule mass	7 000 kg	[TARGET]
Exit doors on capsule	4 (2 side + 1 roof + 1 underside)	[TARGET]
Separation objective	≤ 6 s to capsule-clear state	[TARGET / OPEN]

The 22 m rotor yields a weight-based disk loading of approximately 60.5 kg/m² at 23 000 kg. This is a sizing indicator, not a guarantee of hover, climb, OEI or hot/high performance. Preliminary ideal hover power is approximately 3.29 MW; with figure-of-merit 0.75 approximately 4.39 MW rotor power is indicated. These are study calculations only.

4. AIRCRAFT GEOMETRY AND CONFIGURATION

Item	Definition	Status
Overall length	Approx. 17.0–18.5 m	[TARGET / STUDY]
Fuselage width	Approx. 4.2–4.7 m	[TARGET / STUDY]
Overall height	Approx. 5.0–5.5 m	[TARGET / STUDY]
Main rotor diameter	22.0 m	[TARGET]
Tail rotor diameter	3.6 m	[TARGET]
Capsule envelope	5.6 × 2.9 × 2.2 m	[TARGET]
Landing supports	8 fixed short mechanical supports	[TARGET]
Retractable gear	None	[LOCKED CONCEPT]
Capsule position	Central occupied boundary	[LOCKED CONCEPT]

The eight landing supports form part of the capsule impact-attenuation system. Load path: ground contact → energy absorber → lower capsule structure → primary structure → occupant restraint. Structural sizing shall consider vertical, longitudinal, lateral and torsional loads, slope, uneven terrain, soft soil, asymmetric touchdown and failed-leg cases.

5. PRELIMINARY MASS ACCOUNTING — COMPLETE AIRCRAFT (23 000 kg)

The following accounting model closes to 23 000 kg. It is a planning model, not a verified mass properties report.

Mass Group	kg	Status
Survival capsule, complete	7 000	[TARGET]
Main airframe excluding capsule	4 400	[ASSUMPTION]
Main rotor, hub and controls	1 800	[ASSUMPTION]
Tail rotor, gearbox and drive	400	[ASSUMPTION]
Engines (two turboshaft)	1 600	[ASSUMPTION]
Transmission, gearboxes and mounts	1 100	[ASSUMPTION]
Fuel system hardware	350	[ASSUMPTION]
Usable fuel allocation	2 550	[ASSUMPTION]
Rapid separation system (airframe side)	900	[ASSUMPTION]
Avionics and electrical	750	[ASSUMPTION]

Mass Group	kg	Status
Hydraulic system	280	[ASSUMPTION]
Thermal / environmental control	220	[ASSUMPTION]
Flight control computers and actuators	250	[ASSUMPTION]
Operating fluids and consumables	300	[ASSUMPTION]
Growth / configuration reserve	1 100	[RESERVE]
TOTAL	23 000	[CALC]

6. OCCUPIED SURVIVAL CAPSULE — DETAILED MASS BREAKDOWN (7 000 kg)

The complete survival capsule mass target is **7 000 kg**. The following bottom-up planning breakdown sums exactly to 7 000 kg.

Capsule Component	kg	Status
Primary CFRP roll-cage + titanium nodes	1 950	[TARGET]
External shell / fairing + thermal protection	500	[ASSUMPTION]
Eight landing supports + energy absorbers	850	[TARGET]
Energy-absorbing seats + restraints (10–12)	680	[TARGET]
Four emergency exit doors + frames + hardware	420	[TARGET]
Parachute system (8-canopy reference)	540	[STUDY]
Internal life-support and medical	190	[ASSUMPTION]
Separation interfaces (capsule-side)	320	[ASSUMPTION]
Avionics, lighting, communication, locator	230	[ASSUMPTION]
Internal secondary structure and padding	420	[ASSUMPTION]
Contingency / growth allowance	300	[RESERVE]
TOTAL CAPSULE	7 000	[TARGET]

Four emergency exit doors: two side doors (port and starboard), one roof hatch, one underside hatch. Each door shall support rapid opening from inside and outside and shall not compromise the primary roll-cage load path.

Illustrative impact energies for the 7 000 kg capsule: 126 kJ at 6 m/s, 224 kJ at 8 m/s, 350 kJ at 10 m/s. These are sizing illustrations only, not allowable values.

7. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

Eight short, thick, fixed mechanical supports located below the survival capsule form part of the impact-attenuation system and remain with the occupants after separation. They provide load distribution and redundancy. Each support shall include defined mechanical energy absorption.

Asymmetric landing with failed supports is a required analysis case.

8. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

Layer	Function
L1	Primary automatic emergency detection and initiation
L2	Independent backup initiation
L3	Independent manual / crew authority
L4	Rapid selective physical separation
L5	Direct-impact survival (structure, seats, supports)
L6	Recovery and post-event survival

Independence shall be assessed for power, signal routing, sensors, processors, physical location, structural vulnerability, fire, EMI/lightning and common-cause failures. AI may assist detection; deterministic safety logic authorizes actuation.

9. DETERMINISTIC SAFETY KERNEL

The Safety Kernel evaluates altitude, airspeed, attitude/rates, rotor state, fire/fuel state, structural state, capsule attachment health, separation clearance, parachute envelope, landing-support availability, sensor validity, power availability and crew authority. No AI or predictive model shall bypass the Safety Kernel or human emergency authority.

10. CONTROLLED RAPID SEPARATION SYSTEM

Selective, staged, controlled mechanism for reducing mass, fire load and energetic hardware associated with the capsule. Components are designated jettisonable only after structural, dynamic, fire, control, trajectory and collision analysis. Candidate components: engines, selected transmission, tail rotor, primary fuel tanks (after isolation), selected external panels. Main rotor release is not a default action. Mandatory protections include mechanical retention, electrical inhibits, independent command paths, arm/fire sequence, release-state sensing, fuel isolation, partial-release detection and capsule-clear verification.

11. PRELIMINARY SIX-SECOND SURVIVAL SEQUENCE

Time	Action
0.0–0.4 s	Validate state; arm release; isolate fuel/energy
0.4–2.2 s	Stage primary high-mass/rotating-system release
2.2–3.8 s	Release tail-rotor/drive and primary fuel hazards
3.8–5.4 s	Release remaining airframe/capsule attachments
5.4–6.0 s	Verify capsule clearance
After clearance	Authorize capsule recovery system

Partial and asymmetric separation are first-class emergency states. The target remains [OPEN] until mechanism tests and simulations demonstrate achievability.

12. MULTI-MODE RECOVERY ARCHITECTURE

Layered recovery: controlled flight recovery, autorotation, selective mass/hazard separation, capsule separation, distributed capsule parachutes (8-canopy study), eight energy-absorbing supports, optional residual-airframe parachute, post-landing survival (locator, communications, lighting, emergency power, rescue interface).

13. EMERGENCY STATE MACHINE

NORMAL FLIGHT → ANOMALY DETECTED → CONFIRMATION → IMPACT ASSESSMENT → RECOVERY ATTEMPT → SURVIVAL AUTHORIZATION → STAGED SEPARATION → CAPSULE CLEARANCE → RECOVERY DEPLOYMENT → DESCENT MANAGEMENT → LANDING ATTENUATION → POST-LANDING SURVIVAL → AUDIT.

14. INDEPENDENCE AND FAILURE MODES

Explicit design cases: unintended separation, partial separation, asymmetric separation, release mechanism failure, sensor loss, automatic detection failure, power-bus failure, communication loss, AI/model failure, fire during separation, released-component collision, parachute outside envelope, landing-support failure, common-cause EMI/lightning/fire/structural damage.

15. PRINCIPAL SYSTEM INTERFACES

CAP-AF (capsule-airframe), CAP-PAR (parachute), CAP-LG (landing supports), CAP-DOOR (four exit doors), JET-PWR, JET-CMD, JET-SNS, FUEL-ISO, REC-LOC, AV-CTRL. All survival-boundary interfaces require Interface Control Documents and common-cause assessment.

16. VERIFICATION AND EVIDENCE GATES

G0 Architecture Freeze; G1 Analytical Closure; G2 Subsystem Demonstration; G3 Integrated Ground Demonstration; G4 Dynamic/6-DOF Simulation; G5 Structural/Crash Testing; G6 Recovery System Testing; G7 Flight Development; G8 Certification/Qualification.

17. CRITICAL OPEN ITEMS

Close mass model; close capsule geometry; size rotor and transmission; close tail rotor; select engines and OEI; develop 1000 km fuel model; develop 6-DOF model; define jettisonable set; prove separation trajectories; develop command architecture; define Safety Kernel; size

parachutes; size landing supports; design four doors; perform crash analyses; define injury criteria; define emergency power; define fire isolation; establish certification basis.

18. DESIGN PHILOSOPHY

Controlled survival, not a single device. Chain: PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE → VERIFY CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT → SURVIVE/LOCATE/RESCUE → AUDIT. This Document 00 is a controlled conceptual baseline, not a certified specification.

END OF DOCUMENT 00 — HSC GENERATION 1 — DOUBLE FONT SIZE VERSION

*Founder: Amirbahador Atai | Capsule 7 000 kg with full component table | Aircraft 23 000 kg with full mass table |
Four exit doors | No empty pages*